Polynomials and polynomial functions

We are interested in solving equations of the form $f = 0$ where f is a polynomial with integer coefficients. This means that f is of the form $f = \sum_{|\mathbf{m}|\leq d} a_{\mathbf{m}}\mathbf{x}^{\mathbf{m}}$ where:

- $\mathbf{m} = (m_1, \ldots, m_n)$ is an n-tuple of non-negative integers.
- $\mathbf{x} = (x_1, \ldots, x_n)$ is an n-tuple of variables.
- $\mathbf{x}^{\mathbf{m}} = x_1^{m_1} \cdots x_n^{m_n}$ is a *monomial* in the variables.
- $a_{\mathbf{m}}$ is the *coefficient* of this monomial in f which is an integer.
- $|\mathbf{m}| = m_1 + \cdots + m_n$ is the *total degree* of this monomial.
- d is an upper bound on the degree of f.

The usual ways of adding and multiplying polynomials makes them into a commutative ring $\mathbb{Z}[x_1, \ldots, x_n]$. The constant polynomial 0 is the additive identity and the constant polynomial 1 is the multiplicative identity in this ring.

Given a commutative ring R with identity, there is a natural ring homomorphism $\mathbb{Z} \to R$ which sends 1 to 1. We will denote the image of an integer a under this homomorphism by a with the understanding that two distinct integers a and b may *become* identical in R.

Given a commutative ring R with identity, we get a natural ring homomorphism $\mathbb{Z}[x_1, \ldots, x_n] \to R[x_1, \ldots, x_n]$ where we take the above natural map on *coefficients* and variables remain unchanged.

Polynomial functions

Giving a ring homomorphism $\mathbf{u} : R[x_1, \ldots, x_n] \to R$ is the *same* as choosing $u_1, \ldots, u_n$ in R such that $\phi(x_i) = u_i$ for each $i = 1, \ldots, n$. Thus, such ring homomorphisms can be *identified* with elements $\mathbf{u}$ of the set R^n.

The *image* $\mathbf{u}(f)$ of a polynomial $f \in R[x_1, \ldots, x_n]$ is the result

$$\mathbf{u}(f) = f(\mathbf{u}) = f(u_1, \ldots, u_n) = \sum_{|\mathbf{m}|\leq d} a_{\mathbf{m}} u_1^{m_1} \cdots u_n^{m_n}$$

of *evaluating* the polynomial f at the n-tuple $(u_1, \ldots, u_n)$. Thus, $\mathbf{u} \mapsto f(\mathbf{u})$ gives a function $R^n \to R$.

We say that $\mathbf{u} = (u_1, \ldots, u_n)$ *solves* (or *satisfies*) *the equation* $f = 0$, if $f(\mathbf{u}) = 0$. Equivalently, we say that $\mathbf{u}$ is a *zero* of the polynomial f when this identity holds in R.

The collection $\mathrm{Map}(R^n, R)$ of *all* functions $R^n \to R$ is a ring under *pointwise* addition and multiplication of functions as follows. Given $F : R^n \to R$ and $G : R^n \to R$, we define:

- $(F + G)(\mathbf{u}) = F(\mathbf{u}) + G(\mathbf{u})$ as elements in R
- $(F \cdot G)(\mathbf{u}) = F(\mathbf{u}) \cdot G(\mathbf{u})$ as elements in R.

One easily sees that the above evaluation of polynomials gives a ring homomorphism $R[x_1, \ldots, x_n] \to \mathrm{Map}(R^n, R)$. The *image* of this is called the ring of polynomial functions on R.

Integral domains

A commutative ring R with identity is said to be an *integral domain* if the product of two elements of R is 0 in R if and only one of them is 0.

Using the Binomial theorem, we write

$$(x_1 + x_2)^k = \sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k-r}$$

in the ring $\mathbb{Z}[x_1, x_2]$, where $\binom{k}{r}$ are integers. It follows that the same identity holds in $R[x_1, x_2]$ for *any* commutative ring R with identity.

Using the substitution $x_1 \mapsto (x-u)$ and $x_2 \mapsto u$, we obtain a ring homomorphism $R[x_1, x_2] \to R[x]$. This gives the identity in $R[x]$

$$x^k = ((x-u)+u)^k = \sum_{r=0}^{k} \binom{k}{r} (x-u)^r u^{k-r}$$

We thus, obtain the identity

$$x^k - u^k = \sum_{r=1}^{k} \binom{k}{r} (x-u)^r u^{k-r} = (x-u)P_k(x)$$

for a suitable polynomial $P_k(x)$ in $R[x]$ of degree at most $k-1$.

Given a polynomial $f(x) = a_n x^n + \cdots + a_0$ in $R[x]$, such that $f(u) = 0$ for some u in R. We write

$$f(x) = f(x) - f(u) = a_n(x^n - u^n) + \cdots + a_1(x-u) = (x-u) \cdot \sum_{k=1}^{n} a_k P_k(x) = (x-u) f_1(x)$$

for some polynomial $f_1(x)$ in $R[x]$. Note that the degree of $f_1(x)$ is at most $n-1$.

> If $f(x)$ is a polynomial of degree at most n such that $f(u) = 0$, then $f(x) = (x-u)f_1(x)$ where $f_1(x)$ is a polynomial of degree at most $n-1$.

Now, if R is an integral domain and $f(v) = 0$ where $u \neq v$, then $0 = f(v) = (v-u)f_1(v)$. Since $v - u \neq 0$, this means that $f_1(v) = 0$. It follows that if R is an integral domain and $u_1, \ldots, u_n$ are n *distinct* zeros, then, by induction, we can write $f(x) = (x-u_1) \cdots (x-u_n) f_n(x)$ where $f_n(x)$ has degree 0! This means that $f_n(x) = a_n$ is an element of R inside $R[x]$. Now, if u is an element of R

which is distinct from each u_i, then $u-u_i \neq 0$ for each $i = 1, \ldots, n$. Since R is an integral domain, the only way for $f(u) = 0$ to hold would be if $f_n(x) = a_n = 0$, which would mean $f(x) = 0$.

> If $f(x)$ is a non-zero polynomial of degree at most n with coefficients in an integral domain R, then it has at most n zeroes.

In other words, if R is an *infinite* integral domain, the only polynomial in $R[x]$ that has *all* elements of R as zeroes is the 0 polynomial. Put differently, $R[x] \to \mathrm{Map}(R, R)$ is a one-to-one homomorphism.

We can identify $R[x_1, \ldots, x_n]$ as the ring of polynomials in a *single* variable x_n with coefficients in the ring $R[x_1, \ldots, x_{n-1}]$ of polynomials in the remaining variables. Moreover, if R is an integral domain, then so is $R[x_1, \ldots, x_n]$. Thus, one can argue by induction to show:

> If R is an infinite integral domain, then $R[x_1, \ldots, x_n] \to \mathrm{Map}(R^n, R)$ is a one-to-one ring homomorphism.

In summary, when R is an infinite integral domain such as the ring of integers $\mathbb{Z}$, the field of rational numbers $\mathbb{Q}$, and so on, we can think of polynomials *as* polynomial functions.

However, as we shall see below, when working with *finite* rings R (or rings which are not integral domains), we should be careful to distinguish the ring $R[x_1, \ldots, x_n]$ of polynomials from its image in $\mathrm{Map}(R^n, R)$ which is the ring of polynomial functions.

Evaluation in $\mathbb{Z}/\langle p\rangle$

When $R = \mathbb{Z}/\langle p\rangle$ for a prime p, we are looking for n-tuples $(u_1, \ldots, u_n)$ of elements of $\mathbb{Z}/\langle p\rangle$ that satisfy f. One can say, equivalently that one is looking for homomorphisms $\phi : \mathbb{Z}[x_1, \ldots, x_n] \to \mathbb{Z}/\langle p\rangle$ that send f to 0.

Clearly, p goes to 0 under such a homomorphism. Moreover, as we have seen, if u is an element of $\mathbb{Z}/\langle p\rangle$, then $u^p = u$. Thus, we see that $x_i^p - x_i$ goes to 0 under such a homomorphism. It follows that ϕ factors as $\psi \circ \eta$ where

$$\eta : \mathbb{Z}[x_1, \ldots, x_n] \to (\mathbb{Z}/\langle p\rangle)[x_1, \ldots, x_n]/\langle x_1^p - x_1, \ldots, x_n^p - x_n\rangle$$

is the natural homomorphism going modulo the corresponding ideal, and

$$\psi : (\mathbb{Z}/\langle p\rangle)[x_1, \ldots, x_n]/\langle x_1^p - x_1, \ldots, x_n^p - x_n\rangle \to \mathbb{Z}/\langle p\rangle$$

is the homomorphism sending x_i to $u_i = \phi(x_i)$.

Each element of the ring $(\mathbb{Z}/\langle p\rangle)[x_1, \ldots, x_n]/\langle x_1^p - x_1, \ldots, x_n^p - x_n\rangle$ can be *uniquely* written in the form $\sum_{|\mathbf{m}|\leq d} a_{\mathbf{m}}\mathbf{x}^{\mathbf{m}}$ where:

- $0 \leq a_{\mathbf{m}} \leq p - 1$.
- $0 \leq m_i \leq p - 1$.

It follows that there are exactly p^n possibilities for $\mathbf{m}$. Moreover, for each such $\mathbf{m}$, there are exactly p possibilities for $a_{\mathbf{m}}$. Thus, there are exactly p^{p^n} possiblities for elements of the ring $(\mathbb{Z}/\langle p\rangle)[x_1,\ldots,x_n]/\langle x_1^p - x_1,\ldots,x_n^p - x_n\rangle$.

As seen above, we have a ring homomorphism

$$\tau : (\mathbb{Z}/\langle p\rangle)[x_1,\ldots,x_n]/\langle x_1^p - x_1,\ldots,x_n^p - x_n\rangle \to \mathrm{Map}\left((\mathbb{Z}/\langle p\rangle)^n, \mathbb{Z}/\langle p\rangle\right)$$

given by the evaluation of polynomials f at points $\mathbf{u}$ in $(\mathbb{Z}/\langle p\rangle)^n$. The ring on the right *also* has p^{p^n} elements! So, we can wonder whether this is an isomorphism.

Given a non-zero element u of $\mathbb{Z}/\langle p\rangle$ we have see that $u^{p-1} = 1$. It follows that the polynomial $1-(x-v)^{p-1}$ takes the value 0 when we substitute x by *any* element $u \neq v$ and 1 when we substitute x by v. Now, suppose f is a po

More generally, given any $\mathbf{u}$ in $(\mathbb{Z}/\langle p\rangle)^n$, we see easily that the polynomial

$$\delta_{\mathbf{u}}(\mathbf{x}) = \prod_{i=1}^{n}\left(1-(x_i-u_i)^{p-1}\right)$$

has the property that it evaluates to 0 at all points $\mathbf{v} \neq \mathbf{u}$ of $(\mathbb{Z}/\langle p\rangle)^n$, and it evaluates to 1 at $\mathbf{u}$.

For later reference, we note that $\delta_{\mathbf{u}}(\mathbf{x})$ has degree $p-1$ in *each* variable, has total degree $n(p-1)$, and the only term of degree $n(p-1)$ is $(-1)^n(x_1\cdots x_n)^{p-1}$.

Given a function F in $\mathrm{Map}\left((\mathbb{Z}/\langle p\rangle)^n, \mathbb{Z}/\langle p\rangle\right)$, we consider the polynomial

$$P_F(x) = \sum_{\mathbf{u}} F(\mathbf{u})\delta_{\mathbf{u}}(\mathbf{x})$$

We see that P_F gives the *same* function as F under the above ring homomorphism τ. In other words, τ is a surjective (onto) ring homomorphism. Since both rings have the same number of elements, we have an isomorphism.

> Every function on $(\mathbb{Z}/\langle p\rangle)^n$ with values in $\mathbb{Z}/\langle p\rangle$ is represeted by a *unique* polynomial function whose degree in each variable is at most $p-1$.

Chevalley-Warning theorems

Given f a polynomial in n-variables, consider the polynomial $g = f^{p-1}$. At any point $\mathbf{u} = (u_1,\ldots,u_n)$ in $(\mathbb{Z}/\langle p\rangle)^n$, if $f(\mathbf{u}) \neq 0$, then $g(\mathbf{u}) = 1$.

As seen above the image $\overline{g}$ of g in $(\mathbb{Z}/\langle p\rangle)[x_1,\ldots,x_n]/\langle x_1^p - x_1,\ldots,x_n^p - x_n\rangle$ can be written as

$$\overline{g} = \sum_{\mathbf{u}} g(\mathbf{u})\delta_{\mathbf{u}}(\mathbf{x})$$

which is a polynomial of degree at most $n(p-1)$. Moreover, the *only* term of degree $n(p-1)$ in $\overline{g}$ is $c = (-1)^n(x_1\cdots x_n)^{p-1}\sum_{\mathbf{u}} g(\mathbf{u})$.

Since $g(\mathbf{u}) = 1$ whenever $f(\mathbf{u}) \neq 0$, we see that this coefficient c is the reduction modulo p of the *number* of non-zero values of f evaluated at points of $(\mathbb{Z}/\langle p \rangle)^n$. Since this set has p^n points, we see that the $(-c) \mod p$ is the number of *zeroes* of f in this set.

In particular, consider the case when the total degree of f is *less* than n, the number of variables. In that case, g has degree *less* than $n(p-1)$ and so also $\overline{g}$. This means that $c = 0$. In this case, we see that the number of zeroes of f in the set $(\mathbb{Z}/\langle p \rangle)^n$ is *divisible by p*.

> If an integer polynomial f in n-variables has degree less than n, then the number of its zeroes in $(\mathbb{Z}/\langle p \rangle)^n$ is divisible by p.

Since p is a prime we have $p \geq 2$.

> If an integer polynomial f in n-variables has degree less than n and it has at least one zero, then it has at least 2 zeroes in $(\mathbb{Z}/\langle p \rangle)^n$.

If f only has monomials of a fixed degree d, we say f is a *homogeneous* polynomial. Classically, homogeneous polynomials of degree at least 1 are called *forms*; a homogeneous polynomial of degree 1 is called a *linear form*, one of degree 2 is called a *quadratic form*, one of degree 3 is called a *cubic form*, and so on.

A homogeneous polynomial of degree at least 1 always has a zero at $(0, \ldots, 0)$. Any *other* zero of such a polynomial is called a non-trivial zero. Applying the above, we obtain:

> If a homogeneous polynomial in n variables has degree at least 1 and less than n, then it has a non-trivial zero in $(\mathbb{Z}/\langle p \rangle)^n$.

These last three highlighted statements are collectively known as Chevalley-Warning theorems.

As a particular consequence, note that a quadratic form in at least 3 variables (i.e. $n \geq 3$) has a non-trivial zero in $(\mathbb{Z}/\langle p \rangle)^n$.